1 New Approach

We selected YOLO11s as the second model to compare with the YOLO11n baseline for personal protective equipment detection. YOLO11s is a larger model with approximately 9.4 million parameters and 21.4 GFLOPs. Both models detect the same 11 classes using the Construction-PPE dataset.

2 Motivation

The main goal is to determine whether the larger YOLO11s model provides better detection accuracy than YOLO11n and whether any improvement justifies the additional computational cost. We expect the larger model to improve the detection of difficult and less frequent PPE classes, although this improvement is not guaranteed.

We also compare precision, recall, and inference latency to evaluate the trade-off between detection accuracy and computational efficiency.

3 Implementation

We fine-tuned the pretrained YOLO11s model using the same dataset split as the baseline: 1,265 training images, 73 validation images, and 78 test images.

Training used 50 epochs, batch size 16, SGD with an initial learning rate of 0.01, momentum 0.937, weight decay 0.0005, three warm-up epochs, and cosine learning-rate decay.

Three runs were performed using random seeds 42, 43, and 44 on an NVIDIA Tesla T4 GPU using Ultralytics 8.4.182. Checkpoints were selected using validation mAP@0.50:0.95. The held-out test set was not used for model selection.

4 Initial results

Table 1 compares the validation performance of YOLO11s with the YOLO11n baseline. YOLO11s results are reported as the mean and sample standard deviation across three runs.

Table 1: Validation performance of YOLO11n and YOLO11s (mean ± standard deviation).

Metric	**YOLO11n**	**YOLO11s**
mAP@0.50:0.95	0.1951 ± 0.0066	0.2062 ± 0.0097
mAP@0.50	0.4760 ± 0.0185	0.4851 ± 0.0286
Precision	0.5807 ± 0.0579	0.5374 ± 0.0580
Recall	0.4661 ± 0.0163	0.4880 ± 0.0071
F1-score	0.5156 ± 0.0140	0.5107 ± 0.0284

YOLO11s improved mean mAP@0.50:0.95 from 0.1951 to 0.2062, an absolute increase of 0.0111, or approximately 5.7% relative to the baseline. Mean mAP@0.50 also increased from 0.4760 to 0.4851.

Recall increased, while precision decreased, indicating a trade-off between detecting more objects and avoiding false positives.

Performance varied across classes. Person detection was relatively strong, with AP@0.50 between 0.864 and 0.879 across the three runs. Helmet AP@0.50 ranged from 0.717 to 0.767.

Results for rare classes were less stable. The validation set contains only eight goggles instances and four no_boots instances, making their per-class estimates particularly uncertain.

The preliminary validation logs reported approximately 9.5 ms of inference time per image on the Tesla T4. This is not yet a controlled median-latency benchmark. A direct speed comparison with YOLO11n requires measurements under comparable hardware and evaluation conditions.